

# Thermal stability enhancement and prediction by ANN model

Ziyu Liu<sup>b</sup>, Xiaoyi Yang<sup>a,\*</sup>

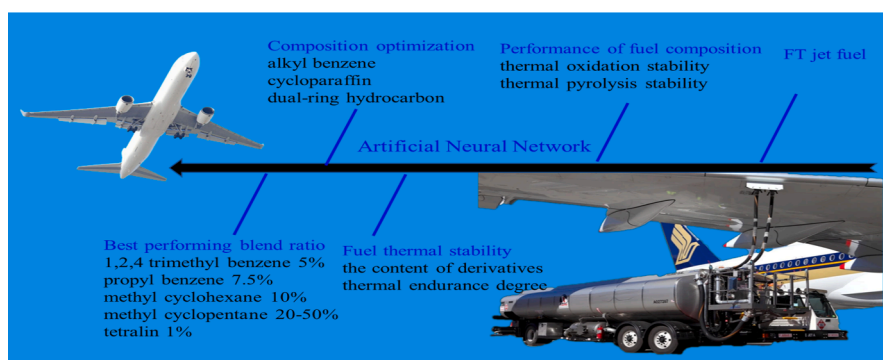
<sup>a</sup> School of Energy and Power Engineering, Energy and Environment International Center, Beihang University, Beijing, 100191, China

<sup>b</sup> School of Aeronautic Science and Engineering, Beihang University, Beijing, 100191, China

## HIGHLIGHTS

- Detail understanding mechanism of thermal stability by blend various additives.
- Discovering structure and pathway of derivatives in thermal endurance.
- Establishing ANN - thermal stability to predict derivatives and the degree of thermal endurance.

## GRAPHICAL ABSTRACT



## ARTICLE INFO

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## ABSTRACT

Supersonic aircraft requires thermal endurance of aviation fuel in the process of cooling engine and aircraft. As the composition of petroleum-based jet fuel (RP-3) is confined by crude oil and refining process, sustainable alternative jet fuel with green house gas reduction become to undertake the composition optimization for improving thermal stability. For designing aviation fuel with robust thermal stability and the detail understanding of thermal stability mechanism, RP-3, Fischer-Tropsch fuel, and additives with cyclic structure for absorbing free radical, were investigated thermal stability by modifying different blend ratios under different conditions. Thermal endurance degree was assessed by chroma and deposition tendency. FT blend with cyclic hydrocarbon can improve thermal endurance degree. In compliance with individual optimized blend ratio, the contribution follows methyl cyclopentane > decalin > methyl cyclohexane > tetralin > n-propyl-benzene > 1,2,4 trimethyl-benzene. The appropriate blend ratio could undertake hydrogen donors for terminating the propagation of oxygen-carrying radical, but hydrocarbons with cyclic structure could enhance deposition tendency. Methyl cyclopentane and its oxidation derivatives take the roles of solvent by anti-polymerization and hydrogen donor by opening cyclic structure in the thermal endurance process, and thus lead to a wide range of blend ratio for improving significantly thermal stability.  $\beta$ -scission leading to C-C bond cleavage is the major reaction at the early decomposition stage, which resulted in most abundant derivatives plus C<sub>2</sub>. The effects of additives on thermal stability are complex and nonlinear on the tendency of thermal deposits and thermal endurance degree, and thus the appropriate ANN-thermal stability model has been trained based on the experiment data and can achieve above 0.995 correlation coefficient. ANN - thermal stability model can predict

\* Corresponding author at: 37 Xueyuan Road, Haidian District, Beijing, 100191, China.

E-mail address: [yangxiaoyi@buaa.edu.cn](mailto:yangxiaoyi@buaa.edu.cn) (X. Yang).

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not only the content of derivatives including ester, olefin, alcohol, ketone, cyclic oxide, aromatics but also the degree of thermal endurance.

## 1. Introduction

Alternative jet fuels blends are confirmed for improving global climate change and maintaining aviation sustainability. In addition, Fischer–Tropsch (FT) process jet fuel could achieve 50–90 wt.% green house gas (GHGs) reduction in life cycle and the potential reduction on particulate matter (PM). Therefore, FT jet fuel attracted extensive research for green aviation. Meanwhile, the composition of petroleum-based jet fuel is confined by its crude oil and refining process, and thus alternative jet fuel becomes to undertake composition optimization for higher level jet fuel. Thermal oxidative stability is one of the key issues of jet fuels for achieving as coolants for high speed of aircraft [1–3]. In addition, thermal oxidative stability of jet fuel is crucial to current commercial aircraft under descent conditions for severe thermal stress due to prolonged residence time for fuel flow reduction [4,5]. For a commercial aircraft, the fuel temperature can be increased 120–160 °C under cruise conditions [6] and the liquid hydrocarbon fuel could be subjected to a temperature over 200 °C at Mach 3+, where the major airframe is exposed to approximately above 300 °C. Therefore, jet fuel with valid thermal stability is of great significance to satisfy the requirements of high speed of aircraft [7]. The thermal oxidative stability of jet fuel attracted much attention in the world [8]. There are some achievements in the thermal stability of jet fuel, including oxidation/deposition mechanisms and improvement methods [7,9].

In the respect of oxidation and deposition mechanisms, fuel stability has been classified as thermal oxidation stability and thermal pyrolysis stability, which is related to the oxidation of jet fuel through the reactions with dissolved oxygen above 150 °C and the formation of gums or deposits by cracking reactions and followed by polymerization above 260 °C [7]. Thermal stability can be mainly divided into two typical regimes according to the characteristics of the products, including the formation of oxidation products by cracking and condensation reaction, and the formation of deposits by pyrolysis and polymerization [11]. Jet fuels usually contain 50–70 ppm dissolved oxygen in open air condition [12]. Above approximately 150 °C, the dissolved oxygen attacks hydrocarbons in jet fuel to form oxygen-carrying free radical species and initiating chain reaction. The propagation of autoxidation reactions could lead to oxidation products formation and fuel destabilization without taking action to ruin free radical species [13,14]. In this regime, oxygen dissolved in fuel is the major cause for fuel destabilization [15,16]. With oxygen depletion to form oxidation products below 5 ppm, the interaction reaction of oxidation products become dominant, which could result in oxidation products cracking into micromolecular oxidation products or condensing into macromolecular oxidation products [17]. Above approximately 260 °C, chemical bonds can be broken to initiate thermal pyrolysis and results in the formation of deposits precursor by polymerization [18]. The fuel may undergo dissociation, cyclization, condensation and finally polymerize into solid deposits, and their deposition rates increase nearly exponentially with temperature [10]. The main components in the deposits are polycyclic aromatic hydrocarbons (PAHs), and thermal oxidative deposits contains 20–30 wt.% oxygen [19]. In the regime of thermal oxidation stability, cycloparaffin and aromatics have been found to prohibit the propagation of oxygen-carrying free radical species as hydrogen donor [20], which can improve the thermal oxidation stability. However, in the regime of thermal pyrolysis stability, cycloparaffin and aromatics are easily to form the deposits of polycyclic aromatic hydrocarbons [21]. Therefore, thermal stability should be considered both of thermal oxidation stability and deposition tendency.

For improving thermal stability of jet fuel, the compositions in jet fuel have been investigated in detail [9,22]. As aviation fuels are

composed of complex hydrocarbons including paraffin, cycloparaffin, aromatics [23], thermal stability was studied based on composition classification. In comparison with thermal oxidation stability of n-paraffins with iso-paraffins, the results confirmed a nonlinear effect of the paraffin branching on thermal stability [24]. The iso-paraffin stability was found to exponentially decrease with the number of branching under the experimental conditions [24]. Similar behavior is known that n-paraffin with longer chain length could decrease the stability [22]. This rule also conforms to alkylcyclohexane which decreases thermal oxidation stability with increasing side-chain length. According to half-life time, the order of thermal stability is methylcyclohexane > ethylcyclohexane > propylcyclohexane followed by less changes with further enhancement side-chain length [25]. Thermal oxidation stability showed a significant relationship with molecular structure [26]. In the range of 120–160 °C, The thermal oxidation stability follow the order: methyl-naphthalene > 1,2,4 trimethyl-benzene > xylene > n-propyl-benzene > n-decane > n-propyl-cyclohexane > decalin > tetralin. They can be ranked as follows alkylaromatics > n-decane > cyclic alkanes > decalin > tetralin. For thermal pyrolysis stability, coking rate decreases in the order naphthalene derivatives > benzene derivative > alkenes > cycloalkanes > alkanes. For paraffins, coking was relatively independent of molecular size, and coking rate order: primary < secondary < tertiary carbon atoms. The coking rate of naphthenes was about 50 wt.% higher than that of paraffins.

In the thermal oxidation stability enhancement, the optimization method is to design jet fuel by modification of jet fuel compositions [27,28]. According to a high thermal-stable jet fuel design method developed by composition-properties relations [29], an optimum chemical composition was established to meet performance requirements: n-paraffins 25–45 wt.%, iso-paraffins 15–30 wt.%, cycloparaffins 30–50 wt.%, and aromatics 5 wt.%. The thermal stability test on four fuel samples with and without optimal composition indicated that - thermal stabilities of fuel samples with optimal composition are higher than that of petroleum jet fuel (RP-3).

A jet fuel with high thermal stability is mainly defined by chemical composition in jet fuel [30,31]. The classification of petroleum-based jet fuel is confined by its crude oil and refining process [32], and thus alternative jet fuels are promising to undertake the composition optimization for thermal stability enhancement. Alternative aviation bio-fuels usually contain only paraffin hydrocarbon with a high thermal pyrolysis stability but a low thermal oxidation stability [33,34]. The addition of decalin and tetralin can improve thermal oxidation stability of alternative jet fuels [26] while alkylbenzene and cycloparaffin have an oxidation inhibiting effect on trapping effectively alkyl radicals.

Artificial neural network (ANN) became a powerful modeling technique with its high reliability and efficiency. The complex problems that are extremely unviable with traditional theory and data analysis can be possibly solved by ANN powerful tool. ANN can bridge a gap for prediction of complex relationship. Neurons in ANN are connected for evaluating complex nonlinear system behavior. For a data subset with a limited case study, principal component analysis has employed for improving the accuracy. In fuel aera, ANN model was developed for analysis of fuel spray performance [35].

For detail understanding the structure of derivatives in thermal endurance and induced composition in jet fuel, hydrocarbon additives with various structures were selected to blend with FT fuel to investigate the effects on thermal stability. FT fuel was chosen as jet fuel substance due to only n-paraffin and iso-paraffin. Additives including alkyl benzene, cycloparaffin, dual-ring hydrocarbon were investigated the derivatives in detail, which can provide insight concerning potential reaction pathways for thermal oxidation and thermal pyrolysis. As the

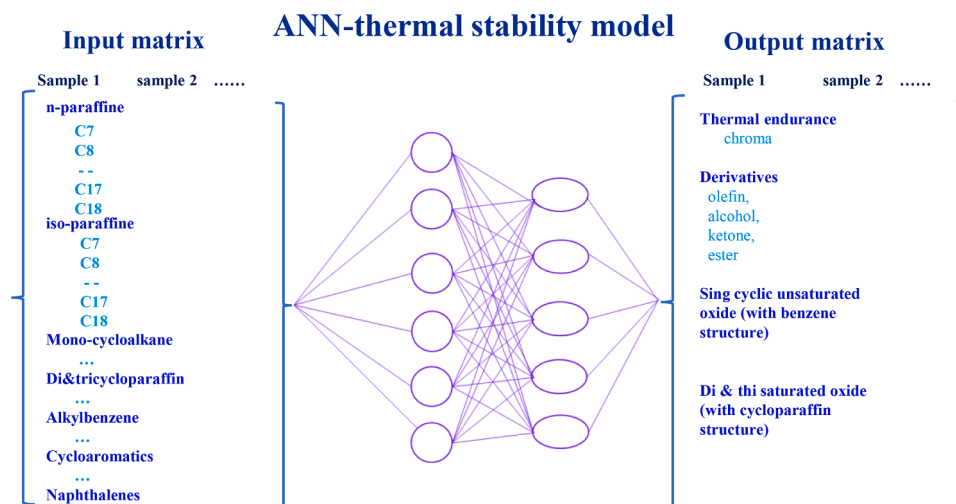


Fig. 1. ANN - thermal stability model frame structure.

research results observe a non-linear and non-monotonous relationship of hydrocarbon composition blend with thermal stability, the artificial neural network (ANN) approach has been involved to solve the complex relationship of compositions of jet fuels with thermal stability. ANN - thermal stability model established can predict derivatives and the degree of thermal endurance with higher accuracy, which can be used to design compositions of jet fuel with high thermal stability.

## 2. Material and methods

### 2.1. Thermal endurance experiments

In order to improve thermal stability and the detail understanding of thermal stability mechanism, additives which can absorb free radicals, were selected for blend with jet fuels, including alkyl benzene (1,2,4 trimethyl-benzene, n-propyl-benzene), cyclopaffin (methyl cyclohexane, methyl cyclopentane), dual-ring hydrocarbon (tetralin, decalin). Besides of thermal stability, alternative aviation fuel should comply with requirement of fluidity, volatility, combustibility, which confined the composition of jet fuel. The range of blend ratio was set according to our previous research in jet fuel composition survey [36,37]. The blend ratios were set in the range of 0.5–50 wt.%.

The thermal endurance experiments were conducted in a stainless-steel autoclave batch reactor with 500 ml, which has a controllable stirrer [38]. In a typical experiment, jet fuel with blend additives (50 ml) were injected into batch reactor and subsequently sealed. The batch reactor was heated to 300 °C keeping for 8 h. After cooling to ambient temperature, jet fuels were collected for investigating the composition of hydrocarbons (classification distribution and carbon number distribution) and derivatives (olefin, alcohol, ketone, ester, cyclic oxide substance) derived from thermal oxidation and thermal pyrolysis.

### 2.2. Compositions of hydrocarbons and derivatives

For discovering compositions of derivatives, derivatives should be extracted first from jet fuel by chromatography method. Fuels collected 20 ml after 8 h thermal endurance at 300 °C, were put in a spherical funnel, and then let jet fuel flow through column with 100–200 meshes neutral alumina. The flow rate was controlled at 20–40 drops /min by a piston valve. Next, the column was washed for removing nonpolar compositions by n-pentane, and then the column was rinsed by anhydrous acetic acid and followed by distilled water. The combined desorption solution was neutralized by 25 wt.% ammonia water and then extracted with methylene dichloride. By a rotary evaporator distillation at 45 °C to remove methylene dichloride, the left residues

were concentrated solution of derivatives, and total weights of which represents the deposition tendency. The concentrated solution of derivatives were diluted with dichloromethane (1:10, V/V) to 2 ml metered volume, then were investigated by GC–MS measurement.

For discovering compositions of hydrocarbons, fuels were diluted with dichloromethane (1:10, V/V) and then were investigated by GC–MS measurement. GC–MS (Agilent 7890A-5975C) system was chosen HP-5 capillary column. The injected volume was 1 µL and the split ratio was 20:1. The oven temperature was held at 50 °C for 2 min, then ramped to 175 °C at 5 °C /min and held for 2 min, and finally ramped to 300 °C at 2 °C /min and held for 2 min. The injector and detector temperatures were 280 and 150 °C.

### 2.3. Chroma measurement

According to the corresponding relationship of total absorbance concentration and chroma at certain wavelength, the chroma of each sample can be measured by establishing the relationship between chroma and absorbance function, which represents the concentration of derivatives in fuel. According to GB/T 6540-1986, 25 kinds of standard colorimetric solution were prepared. The chroma of colorimetric solution was measured by UV-3300 ultraviolet visible spectrophotometer at full wavelength 190–1100 nm with 1 nm scanning accuracy. In order to achieve normalization and sensitivity, the wavelength with stable absorbance rule and the appropriate absorbance within 0.1–1.0 range is selected for chroma calibration. The determined wavelengths are 380 nm (chroma 1–5), 450 nm (chroma 6–13), 550 nm (chroma 14–20) and 620 nm (chroma 21–25). The chroma of the sample is obtained based on the absorbance and chroma curves [38].

### 2.4. ANN-thermal stability model

ANN-thermal stability model was trained by experiment data, given in Fig. 1. The input matrix of ANN - thermal stability model was based on fuel composition at environment temperature, which was classified as n-paraffin, iso-paraffin, alkyl benzene, alkylcyclopentane alkylcyclohexane, dicyclic saturated hydrocarbon, dicyclic unsaturated hydrocarbon. The above critical parameters were extracted based on test data and theory analysis (in Section 3.1).

The output matrix was based on the degree of thermal endurance (chroma) and the derivatives after 8 h the thermal endurance at 300 °C. The derivatives was classified as olefin, alcohol, ketone, ester, sing cyclic saturated oxide (with cyclopaffin structure), sing cyclic unsaturated oxide (with benzene structure), di or thi saturated oxide (with cyclopaffin structure), di or thi unsaturated oxide (with di or thi benzene

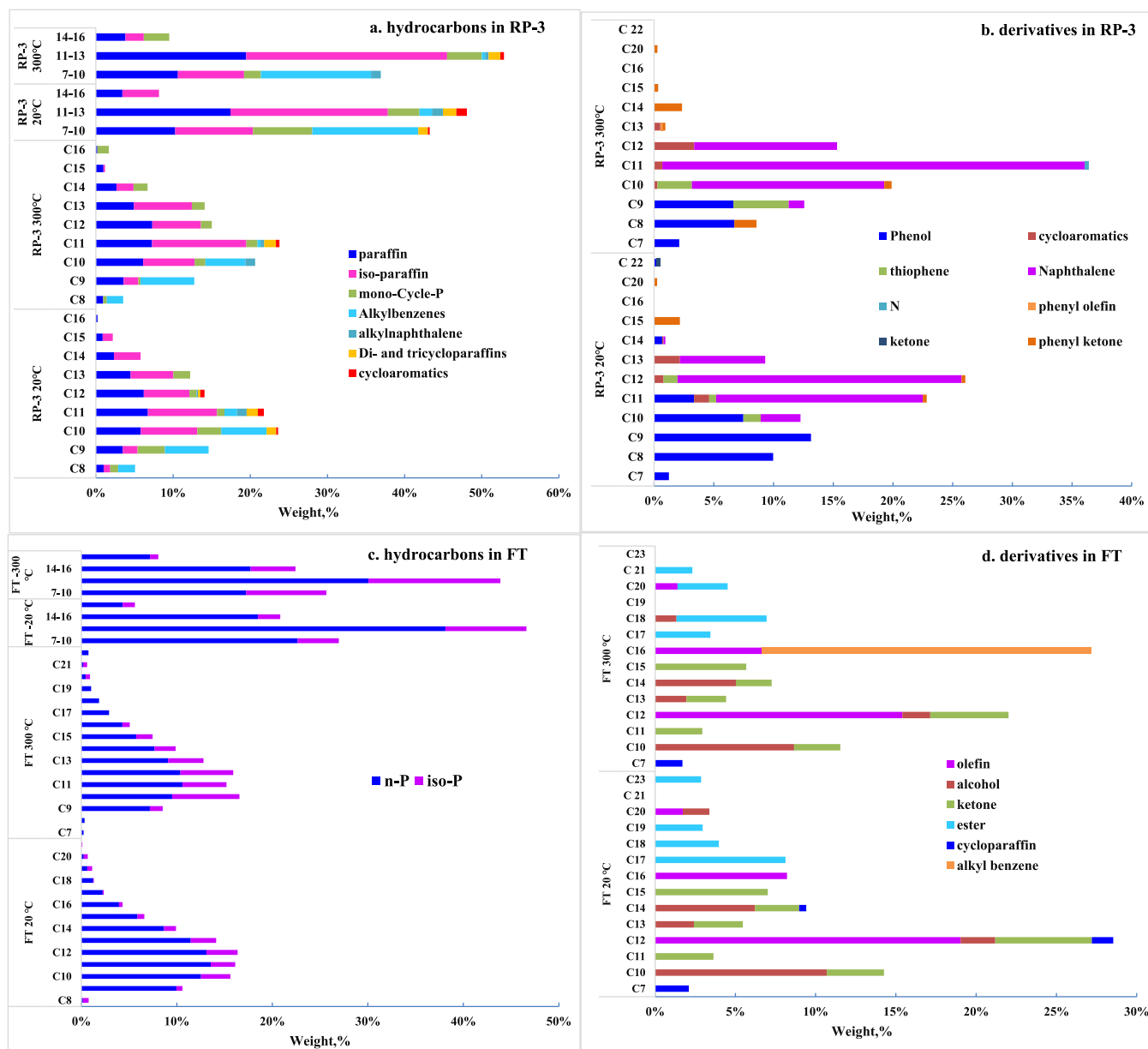


Fig. 2. Comparison of composition changes, a. hydrocarbons in RP-3; b. derivatives in RP-3; c. hydrocarbons in FT; d. derivatives in FT.

ring). The degree of thermal endurance of fuels is involved in training model based on chroma. The deposition tendency depends on derivatives yield. The above critical parameters were extracted based on test data and theory analysis (in Section 3.2).

For achieving the appropriate neural network modeling, 70 wt.% of data set is randomly selected as the training set and 15 wt.% as test set and other 15 wt.% as for validation. ANN model was trained by the back-propagation algorithm with Levenberg–Marquardt rule to minimize the mean square error between test data and predicted outputs.

### 3. Results and discussion

#### 3.1. Storage stability and thermal stability of aviation fuel

The derivatives derived from oxidation or thermal reaction of hydrocarbon in jet fuel, are classified as storage stability derivatives at ambient temperature (10–30 °C, average 20 °C) and thermal stability derivatives at 300 °C. The derivatives provided insight concerning potential reaction pathways for storage stability and thermal stability

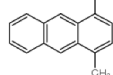
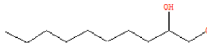
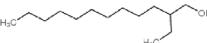
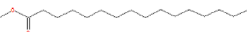
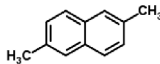
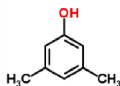
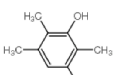
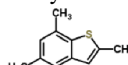
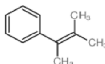
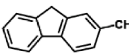
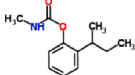
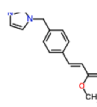
##### 3.1.1. Storage stability

For storage stability at ambient temperature, RP-3 and FT are compared in hydrocarbon compositions and derivative compositions.

RP-3 is relatively more diversiform than FT in classification distribution. RP-3 includes n-paraffin, iso-paraffin, cycloparaffins, alkyl benzene, dicyclic saturated & unsaturated hydrocarbon. The total paraffins of RP-3 sample are 66.6 wt.% (31.1 wt.% n-paraffin, 35.5 wt.% iso-paraffin), given in Fig. 2a. Cycloparaffin and alkyl benzene are also more abundant in RP-3 with the contents of 14.8 wt.% and 15.4 wt.%, respectively. Dicyclic saturated and unsaturated hydrocarbon (naphthalene) are both 1.6 wt.%. In carbon number distribution, RP-3 sample is distributed in the range of C8–C16 with a similar normal distribution, mainly concentrated in C9–C13, among which C10 is the center. However, cycloparaffins and aromatics are concentrated in the narrower range than paraffins. Cycloparaffins are in the range of C8–C13 while aromatics are in the range C8–C11.

From the view of derivative formation at storage process at 20 °C, the main derivatives in RP-3 are naphthalene and alkyl phenol, which are even the main derivatives in RP-3 in spite of storage stability or thermal

**Table 1**  
Typical derivatives derived from FT and RP-3 in thermal endurance.

| Jet fuel | Typical oxidation & thermal pyrolysis derivatives                                                                                   |                                                                                                                          | Classification (weight percentage)               |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| FT       | 5-Dodecene<br>                                     | 1-Pentadecene<br>                       | Olefin (23.46 wt.%)                              |
|          | 7-Pentadecanone<br>                                | 5-Dodecanone<br>                       | Ketone (21.07 wt.%)                              |
|          | Phenol, 2,4-bis(1,1-dimethylethyl)-<br>            | Anthracene, 1,4-dimethyl-<br>           | Alkyl benzene (20.53 wt.%)                       |
|          | 1,2-Decanediol<br>                                 | 2-Ethyl-1-dodecanol<br>                 | Alcohol (18.73 wt.%)                             |
|          | Oxalic acid, allyl pentadecyl ester<br>            | Oxalic acid, allyl pentadecyl ester<br> | Ester (14.51 wt.%)                               |
|          | Cyclohexane, methyl-<br>                           | Cyclododecane, ethyl-<br>               | Cycloparaffin (1.71 wt.%)                        |
| RP-3     | Naphthalene, 2,6-dimethyl-<br>                    | Naphthalene, 1-methyl-<br>             | Naphthalene (64.77 wt.%)                         |
|          | Phenol, 3,5-dimethyl-<br>                        | Phenol, 2,3,5,6-tetramethyl-<br>      | Phenol (15.67 wt.%)                              |
|          | Benzo[b]thiophene, 2,5,7-trimethyl-<br>          | Benzo[b]thiophene, 7-ethyl-<br>       | Thiophene (7.53 wt.%)                            |
|          | 4-Hydroxy-2-methylacetophenone<br>               | Trimethylstyrene<br>                  | Ketone, phenyl ketone, phenyl olefin (5.98 wt.%) |
|          | 1,1'-Biphenyl, 3-methyl-<br>                     | 9H-Fluorene, 2-methyl-<br>            | Cycloaromatics (4.34 wt.%)                       |
|          | Phenol, 2-(1-methylpropyl)-, methylcarbamate<br> | Fluorene<br>                          | Organic nitrogen compound (0.86 wt.%)            |

stability, given in Fig. 2b. Alkyl naphthalene oxidation derivatives are mainly composed of methyl or ethyl naphthalene with 51.7 wt.%. The alkyl phenol contains various alkyl phenols with 36.1 wt.%. Small amounts of thiophene compounds were observed and around 3.22 wt.% in storage stability. Nitrogen polymer can not be detected in storage stability.

The composition of FT fuel is mainly composed of n-paraffin and iso-paraffin, and the content of n-paraffin and iso-paraffin is 83.6 wt.% and

16.4 wt.%, respectively. The carbon number of FT fuel is distributed in the range of C8 - C20 with a similar distribution curve as RP-3, and mainly in the range of C9–C16 (94.4 wt.%) concentrated in C9–C13, given in Fig. 2c. For derivative formation of FT fuel at storage process, the main derivatives are derived from paraffin oxidation and follow the order olefin > ketone > alcohol > esters > cycloparaffin oxide, given in Fig. 2d.

According to derivative formation and concentration order to deduce

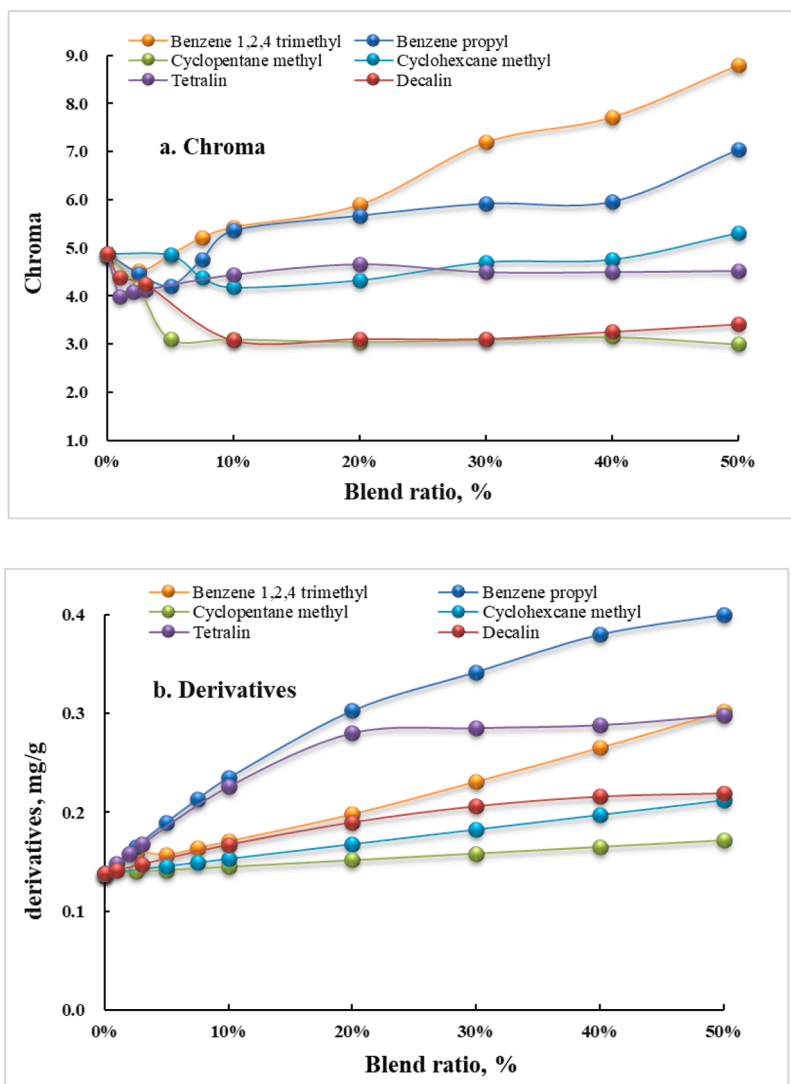


Fig. 3. Thermal endurance degree of FT blend fuels. a. Chroma; b. derivatives weight.

the mechanism, oxidation hydrocarbons as the main derivative yield mainly conduct at oxidation reactions. At storage process, dissolved oxygen (around 50–70 ppm in jet fuel in open air condition) in jet fuel could be induced to attack hydrocarbons. The formation of olefins were usually discovered at  $\beta$  sites of paraffins by hydrogen abstraction while the formation of ketones were derived from  $\beta$  sites of paraffins by oxidation. As olefin concentration is beyond ketone concentration, it can infer that hydrogen abstraction at  $\beta$  site has a higher priority than oxidation dehydrogenation at  $\beta$  site. Integrating with steric hindrance and bond energy,  $\alpha$  site in the carbon chain can undergo oxidation dehydrogenation reaction to form alcohols and aldehydes, which could further be oxidized into acid, which is easy to combine alcohols to esters.

Hydrocarbons undergo oxidation to form free radical  $R\bullet$ , which can combines with oxygen molecules to form more active free radical peroxide  $ROO\bullet$  as the main intermediate carrier of chain reaction, which gradually enhance the oxidation of hydrocarbons to alcohols, ketones, acids, esters. The alkyl radicals can undergo further decomposition to form lower alkyl radicals and olefins. If it is lack of adequate hydrogen donors for absorbing the alkyl radicals, olefin cyclization can rapidly proceed to form cycloparaffin oxide derivatives.

### 3.1.2. Thermal stability

For thermal stability through thermal endurance at 300 °C for 8 h, RP-3 and FT are compared in hydrocarbon compositions and derivative

compositions, given in Fig. 2a–d. Typical derivatives derived from FT and RP-3 in thermal endurance, were given in Table 1.

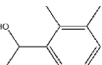
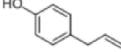
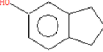
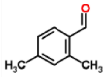
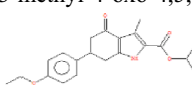
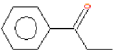
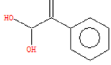
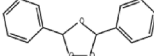
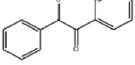
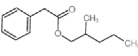
In carbon number distribution, hydrocarbons in RP-3 move to higher carbon number through thermal endurance. Especially, cycloparaffins with C14 and C16 can be detected after thermal endurance, which may be deduced that paraffin was dehydrogenated to form olefin and then followed by cyclization. The concentration of dicycloparaffins and cycloaromatics decreased but the concentration of alkylnaphthalene increased. Dicycloparaffins and cycloaromatics convert to naphthalene as hydrogen donor by absorbing hydroxyl radical. Compared with storage stability, alkyl naphthalene oxidation derivatives are mainly composed of methyl or ethyl naphthalene with 51.7 wt.% at 20 °C and 64.8 wt.% at 300 °C, respectively.

The alkyl phenol contains various alkyl phenols, and the content decrease from 36.1 wt.% at 20 °C to 15.5 wt.% at 300 °C. The unstable alkyl phenols could further polymerize into macromolecular aromatics in thermal endurance process. Small amounts of thiophene compounds were observed and increased from 3.22 wt.% in storage stability to 7.53 wt.% in thermal stability. Nitrogen polymer can be neglect in storage stability but can be captured at 0.35 wt.% in thermal stability, which comply with the previous research that heteroatom could promote further polymerization [7]. After thermal endurance, hydrocarbons in FT also moves to higher carbon number as same as RP-3. In addition, C7 and C27 were captured after thermal endurance at 300 °C, which



**Table 2**

Typical thermal oxidation and pyrolysis derivatives derived from alkyl benzene.

| Additives               | Typical oxidation & thermal pyrolysis derivatives                                                                      |                                                                                                                     |                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,2,4 trimethyl benzene | Benzaldehyde, 2,4-dimethyl-<br>       | Benzenemethanol, 2,5-dimethyl-<br> | 1,2-Benzenedicarboxylic acid, bis(2-methylpropyl) ester<br>                                             |
|                         | Phenol, 4-(2-propenyl)<br>            | 1H-Inden-5-ol, 2,3-dihydro<br>     | Dibutyl phthalate<br>                                                                                    |
|                         | 9,10-Anthracenedione, 1-ethyl-<br>    | Benzaldehyde, 2,4-dimethyl-<br>    | 1H-Indole-2-carboxylic acid, 6-(4-ethoxyphenyl)-3-methyl-4-oxo-4,5,6,7-tetrahydro-, isopropyl ester<br> |
| Propyl benzene          | Benzenecetic acid<br>                 | 1-Phenyl 1-Propanone,<br>          | 1H-Inden-1-one, octahydro-<br>                                                                          |
|                         | Ethanone, 2,2-dihydroxy-1-phenyl-<br> | Benzaldehyde<br>                   | 3,5-Diphenyl-1,2,4-Trioxolane,<br>                                                                       |
|                         | Benzophenone<br>                     | Diphenyl-ethanedione<br>          | Benzenecetic acid, 2-methylpentyl ester<br>                                                             |

indicated both of polymerization and decomposition present in the process of thermal endurance. Especially, n-paraffin decreased from 83.6 wt.% to 72.2 wt.% and iso-paraffin increased from 16.4 wt.% to 27.8 wt.%, which indicated n-paraffin could conduct disproportionated reaction and even isomerization.

The derivatives provided insight concerning potential reaction pathways for thermal oxidation and thermal pyrolysis. In thermal stability, the order derivatives in FT is olefin > ketone > alkyl benzene > alcohol > esters > cycloparaffin. The obvious difference of storage stability and thermal stability is the formation of alkyl benzene in thermal endurance process. The olefin could further polycondensate to form alkyl benzene, and cycloparaffin newly produced could dehydrogenated to alkyl benzene. After the consumption of dissolved oxygen in jet fuel, polymerization and disproportionation take an important role on the propagation of molecular growth.

### 3.2. Derivatives and mechanism analysis in thermal endurance

#### 3.2.1. Thermal oxidation degrees

In order to improve thermal stability and the detail understanding of thermal stability mechanism at molecule structure level, FT fuel was chosen as jet fuel substance due to only n-paraffin and iso-paraffin. Additives with cyclic structure can absorb free radicals, were selected to blend with FT fuel, including alkyl benzene (1,2,4 trimethyl-benzene, n-propyl-benzene), cycloparaffin (methyl cyclohexane, methyl cyclopentane), dual-ring hydrocarbon (tetralin, decalin). From the derivatives of additives in thermal endurance process, the potential reaction pathways for thermal oxidation and thermal pyrolysis can be deduced for mechanism analysis. The key parameters of thermal oxidation degrees and derivatives in thermal endurance are extracted to assess the degree of thermal stability and the deposition tendency,

respectively. The degree of thermal stability concerns the concentration of thermal derivatives and compatibility with jet fuel while the weight of the total derivatives represents the tendency to form deposits. Chroma as thermal oxidation degrees are shown in Fig. 3a and derivatives yield as the deposition tendency in thermal endurance are given in Fig. 3b.

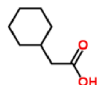
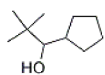
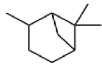
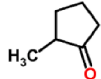
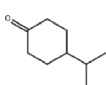
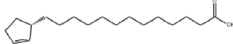
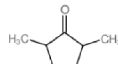
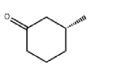
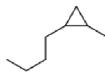
According to the corresponding relationship of total absorbance with chroma at certain wavelength, the chroma represents the concentration of derivatives in fuel and compatibility with jet fuel.

For alkyl benzene performance in thermal stability, both propylbenzene and 1,2, 4-tritylene can decrease chroma at low blend ratio, but then increase significantly at the high blend ratio. The blending ratio of 1,2, 4-tritylene is lower than 5 wt.% and propylbenzene is lower than 7.5 wt.%, the chroma of the blended fuels are lower than that of FT fuel, which indicated that alkyl benzene at low blend ratio can improve the thermal oxidation stability. However, as the blend ratio of propyl benzene and 1,2, 4-tritylene is higher than 10 wt.%, the deep thermal oxidation reaction could occur and the thermal stability of the blend fuel decrease significantly.

For cycloparaffin, the chroma of methyl cyclohexane and methyl cyclopentane decrease significantly with the increase of blend ratio in the range of less than 10 wt.%, and methyl cyclopentane was obviously better than that of methyl cyclohexane in inhibiting the production of thermal oxidation products. Even if the blend ratio methyl cyclopentane reaches at 50 wt.%, it can still effectively inhibit the production of thermal oxidation products, and the chroma is always significantly lower than that of FT fuel. However, the blend ratio of methyl cyclohexane is in the range of 10–40 wt.%, the chroma increases slowly, but it is lower than that of FT fuel. At 50 wt.% blend ratio, the blend of methyl cyclohexane leads to the chroma higher than that of FT fuel. Methyl cyclopentane and methyl cyclohexane were found less derivatives yield than alkyl benzene and dual-ring hydrocarbon.

**Table 3**

Typical thermal oxidation and pyrolysis derivatives derived from cycloparaffins.

| Additives           | Typical oxidation & thermal pyrolysis derivatives                                                                                   |                                                                                                                                   |                                                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Cyclopentane methyl | 1,2,3,5-Cyclohexanetetrol<br>                      | 3-Methyl cyclopentanone<br>                      | Cyclohexaneacetic acid<br>        |
|                     | 1-Cyclopentyl-2,2-dimethyl-1-propanol<br>          | Bicyclo [3.1.1] heptane, 2,6,6-trimethyl<br>     | Cyclopentanone, 2-methyl-<br>     |
|                     | 4-Isopropyl cyclohexanone<br>                      | 2-Cyclopentene-1-undecanoic acid<br>             | Cyclopentanone, 3-methyl-<br>     |
| Cyclohexane methyl  | Cycloheptanol<br>                                  | Cyclohexanone, 2-methyl-<br>                     | Cyclopentanone, 2,5-dimethyl-<br> |
|                     | Cyclohexanone, 3-methyl-<br>                       | Bicyclo[2.2.1]heptan-2-ol,2,3,3-trimethyl-<br>   | 1-Butyl-2-methylcyclopropane<br>  |
|                     | Anthracene, 9-ethyl-9,10-dihydro-10-t-butyl-<br> | Anthracene, 9,10-dihydro-9,9,10-trimethyl-<br> | 2-Acetylcyclopentanone<br>      |

For dual-ring hydrocarbon, the blend of tetralin and decalin have a similar influence on chroma. The chroma can significantly reduce at even lower blend ratio, and then the chroma increase slightly with blend ratio but can still keep lower than FT fuel. At the same blend ratio, decalin always perform better than tetralin. The best blending ratio of tetraline is at 1 wt.% while decaline can improve the overall thermal oxidation stability in a wide range of blend ratios within 10–30 wt.%.

### 3.2.2. Derivatives and mechanism analysis

For the effect of alkyl benzene blend, 1,2,4 trimethyl benzene and propyl benzene were compared in structure of derivatives. Typical oxidation and thermal pyrolysis derivatives are given in Table 2.

Both of alkyl benzene blend fuels cannot captured oxidation product with benzene ring at 2.5 wt.% blend, and only aliphatic compounds were detected, which indicate that alkyl benzene undergo ring open reaction to absorb dissolved oxygen in fuel. At 5 wt.% and 7.5 wt.% blend, the derivatives derived from 1,2,4 trimethyl benzene can show evidence of oxidation products including dimethyl benzaldehyde, dimethyl benzene methanol, phenol, 4-(2-propenyl). Above 10 wt.% blend, bicycle substances and tricycle substance were discovered. Bicyclic substances (1H-Inden-5-ol, 2,3-dihydro) was detected in 10 wt.% blend while biphenyl substances (isopropyl ester 9,10-anthracenedione) and tricycle substance (1-ethyl-1H-Indole-2-carboxylic acid, 6-(4-ethoxyphenyl)-3-methyl-4-oxo-4,5,6,7-tetrahydro-) were detected in 20 wt.% blend. Chroma of blend fuel conduct nonmonotonic relationship with blend ratio. Chroma of FT blend fuel present lower at below 2.5 wt.% blend but present higher at above 5 wt.% blend. However, the derivative yields always keep increasing along with blend ratio. 1,2,4 trimethyl benzene blend below 5 wt.% improve the oxidation stability. The

derivative yield were enhanced at 2.5 wt.% blend ratio while chroma were enhanced at 5 wt.% blend ratio. At lower blend ratio, 1,2,4 trimethyl benzene can achieve the role as sacrificial agent to capture dissolved oxygen in fuel, and the derivatives mainly contain phenols, which have compatibility with fuel. At above blend ratio after dissolved oxygen depletion, 1,2,4 trimethyl benzene could undergo thermal pyrolysis to form polycyclic hydrocarbon such as dibutyl phthalate and anthracenedione.

For propyl benzene, the oxygen-containing free radicals mainly attack its side chain carbon to produce ketone (1-phenyl 1-propanone), ester (benzeneacetic acid, 2-tetradecyl ester), aldehyde (benzaldehyde), alcohol (ethanone, 2,2-dihydroxy-1-phenyl-). As same as 1,2,4 trimethyl benzene, propyl benzene blends also enhance the derivative yield even at 2.5 wt.% blend ratio. Chroma of FT blend below 7.5 wt.% present lower than those of FT fuel. Propyl benzene can enhance the oxidation stability at 7.5 wt.% blend ratio than 5 wt.% 1,2,4 trimethyl benzene. However, the derivative yields of propyl benzene blend, which represent the deposition tendency, are higher than those of trimethyl benzene. At above 10 wt.% blend, double rings and tricycle oxidation products appear in fuel.

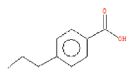
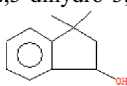
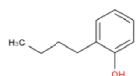
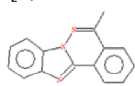
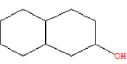
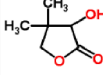
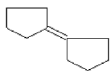
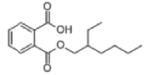
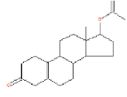
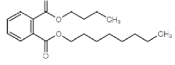
For single-cycloparaffin blend, methyl cyclopentane and methyl cyclohexane were compared in structure of derivatives, given in Table 3.

Methyl cyclohexane blend can be oxidized mainly into methylcyclohexanone, cycloheptanol, isopropyl cyclohexanone. It shows that methyl cyclohexane plays an important role of hydrogen donor in oxidation stability, which absorbs dissolved oxygen to oxidize into cyclohexanone and cyclohexanol and reduce paraffin oxidation, given in Table 3. But above 20 wt.% blend, bicyclic substance (bicyclo[2.2.1]heptan-2-ol, 2,3,3-trimethyl-) and macromolecule triphenyl ring



**Table 4**

Typical thermal oxidation and pyrolysis derivatives derived from dicycle hydrocarbon.

| Additives | Typical oxidation & thermal pyrolysis derivatives                                                                        |                                                                                                                                     |                                                                                                                                               |
|-----------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Tetralin  | 1(2H)-Naphthalenone, 3,4-dihydro<br>    | 4-n-Propylbenzoic acid<br>                         | Indan, 1-methyl-<br>                                        |
|           | 1-Naphthalenol, 1,2,3,4-tetrahydro-<br> | 1H-Inden-1-ol, 2,3-dihydro-3,3-dimethyl-<br>       | Phenol, 2-butyl-<br>                                       |
|           | 1-Naphthalenol, 5,6,7,8-tetrahydro-<br> | Benzo[4,5]imidazo[2,1-a]phthalazine, 5-methyl-<br> | Phenol, 2,4-bis(1,1-dimethylethyl)-<br>                    |
| Decalin   | decahydro-1- Naphthalenol,<br>          | decahydro-2-Naphthalenol<br>                       | 2(3H)-Furanone, dihydro-3-hydroxy-4,4-dimethyl-<br>        |
|           | Bicyclopentylidene<br>                  | 3-tetradecyl ester cyclobutanecarboxylic acid<br>  | 1,2-Benzenedicarboxylic acid, mono(2-ethylhexyl) ester<br> |
|           | Tricyclo[4.4.0.0(2,8)]decane<br>      | Estra-5(10)-en-3-one-17-ol, acetate<br>          | 1,2-Benzenedicarboxylic acid, butyl octyl ester<br>      |

substance (anthracene, 9-ethyl-9,10-dihydro-10-t-butyl-, anthracene, 9,10-dihydro-9,9,10-trimethyl-) were formed which results in chroma deeper.

The derivatives in methyl cyclopentane blend include 1-cyclopentyl-2,2-dimethyl-1-propanol, cyclohexanetetrol and isopropyl cyclohexanone. Methyl cyclopentane can absorb oxygen-carrying free radical to form 1-cyclopentyl-2,2-dimethyl-1-propanol or methyl cyclopentanone. Comparison with methyl cyclohexane, methyl cyclopentane takes the role of hydrogen donor in the thermal endurance process coupling its five-ring structure open to form like isopropyl cyclohexanone and 2,6,6-trimethyl bicyclo[3.1.1]heptane. At 50 wt.% methyl cyclopentane blend, derivatives with cyclopentane structure oxides to aromatics was 1.4:1 while derivatives with cyclohexane structure oxides to aromatics was 12.3:1. The results indicated methyl cyclohexane blend could yield more deposits than methyl cyclopentane. Moreover, because methyl cyclopentane itself and its oxidation derivatives have solvent effect and thus it shows that a wide range of blend ratio can significantly improve thermal stability (Table 3).

For dicycle substance blend, thermal oxidation derivatives in 20 wt.% tetralin blend contain 86.3 wt.% ring structure oxide and 13.7 wt.% chain oxide while those in 40 wt.% tetralin blend contain 88 wt.% oxides with ring structure and 12 wt.% oxidation products with chain structure. Naphthalenol, naphthalenone and indene were detected. It shows that tetralin absorb oxygen-carrying free radical above 80 wt.%. In addition, there are some derivatives with single benzene ring including phenol and phenyl ketone, which indicated tetralin break the bond and open the ring for directly connecting with oxygen-containing free radicals. Below 10 wt.% blend, the absorption of oxygen-containing free radical is greater than polymerization of its own oxidation products, and thus the

total contribution decreases obviously the chroma of blend fuel, and the best performance was observed at 1 wt.% blend. Although it still has the effect of inhibiting the thermal oxidation of chain hydrocarbons at higher blend, the chroma of blend fuel present no obvious change at the range of 20–50 wt.% blend due to the ring-opening and polymerization of its own oxidation products to form macromolecular polycyclic structure substances. Comparison with tetralin, decalin blend present better as hydrogen donor to absorb free radicals in the oxidation reaction. The dehydrogenation and oxidation products of decalin were detected including decahydro-1- naphthalenol, cyclohexane, 1-ethyl-1,4-dimethyl, bicyclopentylidene. Its oxidation products also have good hydrogen supply capacity and absorption ability of free radicals due to with the stable structure of cyclohexane including methyl-cyclohexane, 1-ethyl-1, 4-dimethyl-cis-cyclohexane, ketone, 4-ethyl-3, 4-dimethyl-cyclohexanone, 2-acetyl-cyclohexanone given in Table 4. Thus, decalin has a good effect on improving stability in the process of thermal endurance. Decalin blend perform as same as methyl cyclopentane, showing enhancement of thermal oxidation stability in a wide range of blend.

### 3.3. ANN-thermal stability model performance

The influence of the blending ratio is complex and nonlinear on the tendency of thermal deposit and thermal endurance degree. For predicting the thermal stability of jet fuel and design blend fuel with higher thermal stability, the appropriate ANN-thermal stability model was established based on the experiment data. The appropriate structures and topology of networks make an important role for achieving ANN performance. The multiple-input is related with critical compositions in

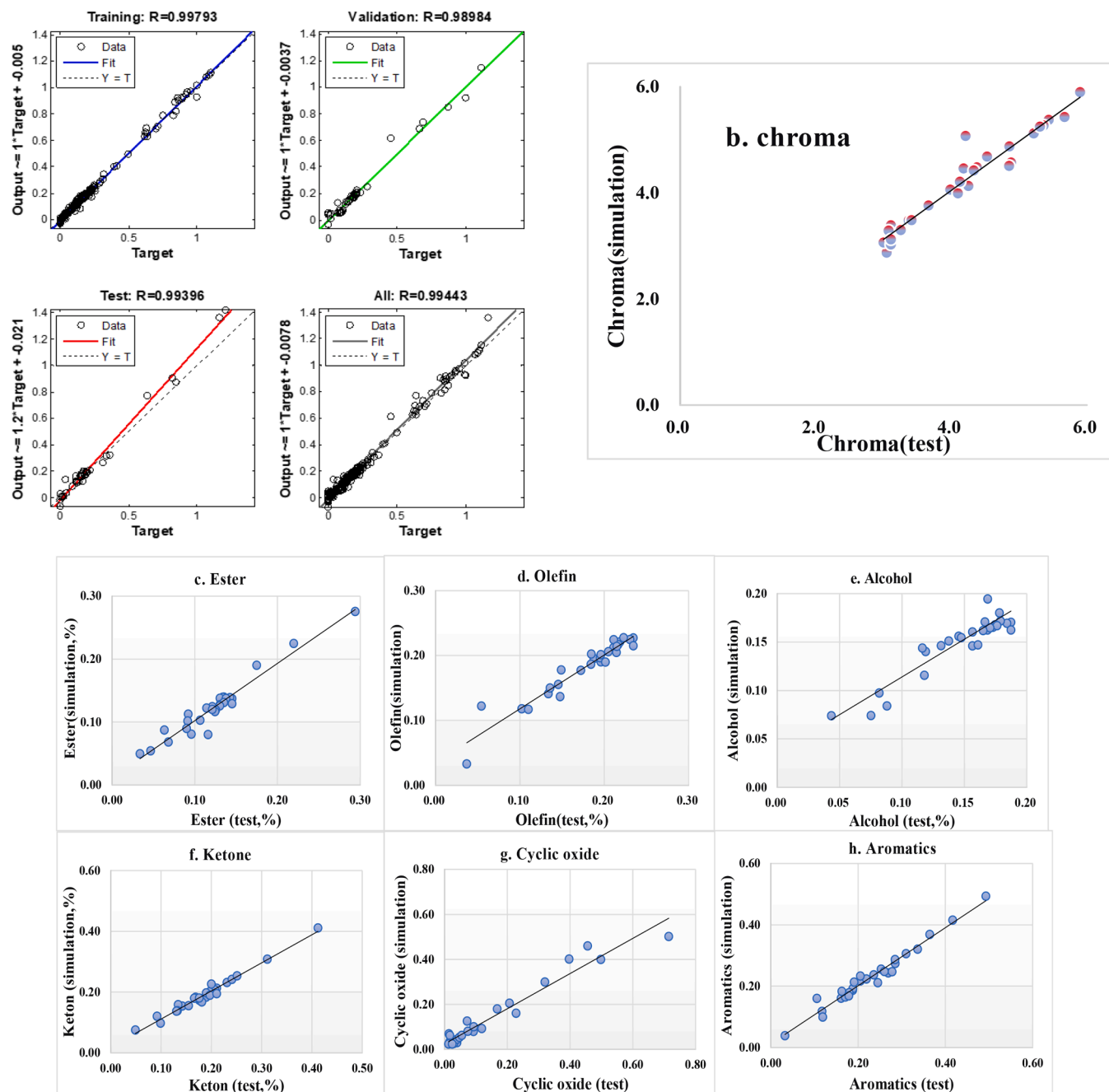


Fig. 4. ANN-thermal stability model performance, a. ANN-thermal stability performance; b. chroma; c. ester; d. olefin; e.alcohol; f. ketone; g. cyclic oxide; h. aromatics.

fuel while multiple-output is related with the degree of thermal endurance and derivatives composition.

From the view of derivatives yield, the order of thermal endurance degree can be ranked as methyl cyclopentane > methyl cyclohexane > decalin > 1,2,4 trimethyl-benzene > tetralin > n-propyl-benzene. Moreover, additives with different structure are quite different in the appropriate blend ratio. Hydrocarbon additives with ring structure could undertake hydrogen donors for termination the propagation of oxygen-carrying radical, but the results indicated that those could enhance the tendency of deposits formation. From the view of chroma, the order of thermal endurance degree are complex and nonmonotonic. 1,2,4 trimethyl benzene blend performs better in thermal stability at 5 wt.% while propyl benzene at 7.5 wt.%, methyl cyclohexane at 10 wt.%, decalin at 10 wt.%, tetralin at 1 wt.%, methyl cyclopentane at (20 wt.

%–50 wt.%), respectively. Additives with cyclic structure can absorb free radicals to inhibit oxidation products but it is easy to form deposits by pyrolysis and polymerization, which should be balanced in thermal stability.

According to derivatives classification and structure, derivatives of paraffin oxidation and pyrolysis contain olefin, ketone, alcohol, and ester while derivatives of cycloparaffin and cycloaromatics contain cyclic oxide and aromatics. The major reaction pathways of cycloparaffin and cycloaromatics are strongly dependent on the side-chain length and cyclic structure.  $\beta$ -scission leading to C–C bond cleavage is the major reaction at the early decomposition stage, the decomposition resulted in most abundant derivatives plus C2. Integrating the derivatives yield and chroma as thermal endurance degree, compositions in fuels undergo complex pathways by oxidation and pyrolysis, and meanwhile

intermediate products could further react to derivatives. The complex relationship of multiple-input (compositions) with multiple-output (derivatives and chroma) are complex and nonmonotonic. Therefore, ANN model was involved to establish the relationship.

ANN model was trained by the back-propagation algorithm with Levenberg-Marquardt rule to minimize the mean square error between test data and predicted outputs. In the domain of neural network modeling by multi-round iteration, the high-precision ANN-thermal stability model was created with 70 wt.% training data at 0.9979 correlation coefficient, 15 wt.% validation at 0.9898 correlation coefficient, and 15 wt.% test at 0.9940 correlation coefficient, and the total at 0.9944 correlation coefficient, given in Fig. 4a.

In Fig. 4b, ANN - thermal stability can predict thermal oxidation degree by chroma. The tiny change in fuel compositions could induce deviation of chroma as thermal endurance degree, which can be captured by model. Moreover, olefin, ketone, alcohol, and ester were derived from paraffin oxidation and pyrolysis and cyclic oxide and aromatics were derived from oxidation and pyrolysis of cycloparaffin and cycloaromatics, all of them can be predict by model, given in Fig. 4c–h.

Although thermal endurance degree are nonmonotonic and derivatives include various substances, ANN - thermal stability can predict well not only the content of derivatives based on classification but also the degree of thermal endurance.

#### 4. Conclusion

Cyclic hydrocarbon can improve thermal endurance degree in compliance with the restriction of blend ratio. The optimum of blend ratio is 5 wt.% 1,2,4 trimethyl benzene, 7.5 wt.% propyl benzene, 10 wt.% methyl cyclohexane at, 10 wt.% decalin, 1 wt.% tetralin at, 20–50 wt.% methyl cyclopentane, respectively. The major reaction pathways of cycloparaffin and cycloaromatics are strongly dependent on the side-chain length and cyclic structure. Decalin has a good effect on improving stability in the process of thermal endurance. Decalin blend perform as same as methyl cyclopentane, showing enhancement of thermal oxidation stability in a wide range of blend.

The alkyl phenols derived from storage stability derivatives could polymerize into alkyl naphthalene in thermal endurance process. Nitrogen polymer can be neglect in storage stability but can be detected in thermal stability.

Blend fuel can be deduced the formation of olefins by  $\beta$  scission from paraffin by hydrogen abstraction, and olefin cyclization can rapidly proceed to form alkyl benzene derivatives. Subsequently, polymerization and disproportionation could further contribute on the propagation of molecular growth.

ANN - thermal oxidation stability can predict not only the content of derivatives based on classification but also the degree of thermal endurance.

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#### CRediT authorship contribution statement

**Ziyu Liu:** Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Xiaoyi Yang:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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